

Programming Fundamentals

Spring 2026



Homework 1

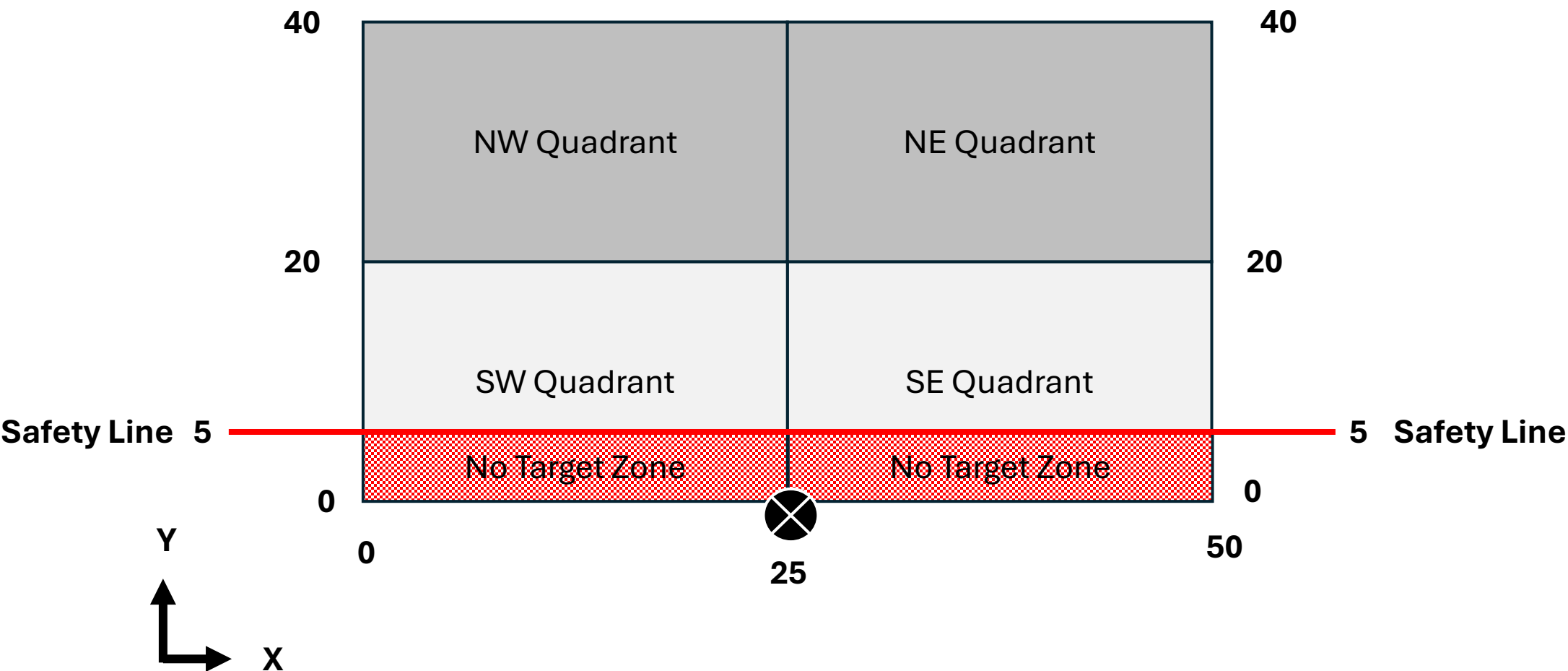
Simple Problem Solving

Professor HG Locklear

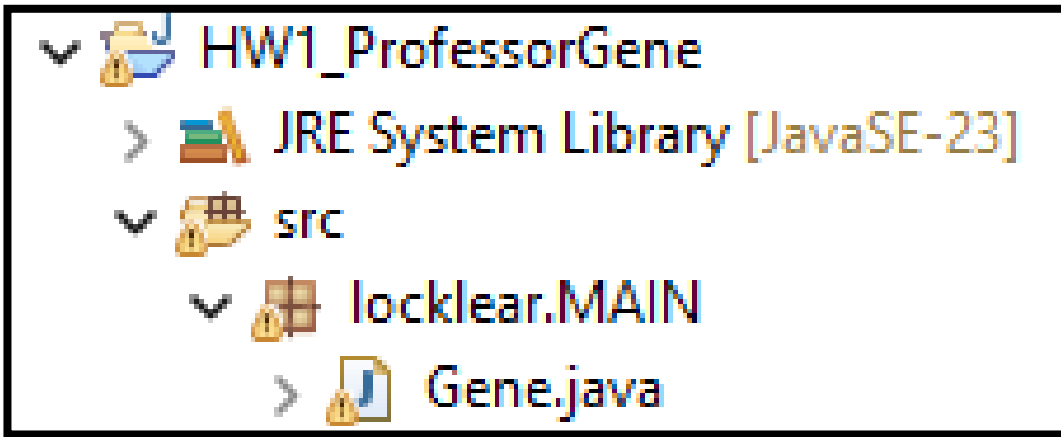
Problem

- Stormtroopers regularly visit the marksmanship range to hone their exceptional marksmanship skills. We have been asked by Imperial Training Command to **create a Java program to simulate this process.**
- **WHAT WE KNOW ABOUT THE RANGE**
- The range is a **50 meter by 40 meter rectangular-shaped box** consisting of a firing line and four delineated quadrants where targets may appear. (See slide 3)
- **A portion of the range is excluded for safety reasons** and no targets may appear there. (See Slide 3)
- The targets **appear randomly**, on the range, in **different quadrants** and at **different distances.**
- The marksmanship range provides the stormtrooper **10 targets** to engage, some of which may be out of range of his weapon.
- A target's distance from the stormtrooper is calculated using 2D Euclidean distance and is classified as **Short** (0-10m inclusive) **Medium** (11-20m inclusive) **Long** (21-30m inclusive) ***OUT*** (> 30m)
- **WHAT WE KNOW ABOUT THE STORMTROOPER**
- The stormtrooper is positioned on the range at **X: 25** and **Y: 0** and has a starting accuracy of **70%.**
- The stormtrooper's **accuracy is decreased by 5%** for the next shot, if he missed on the previous shot.
- The stormtrooper **will not shoot** at targets that are out of range.

Problem Graphic



Program Structure and Requirements



Structure your program **EXACTLY** as shown replacing my name with yours where appropriate.

Program Restrictions

- The location (x, y) of the stormtrooper **must be** represented as static variables **defined as constants**.
- The maximum range of the stormtrooper's weapon **must be** represented by a static variable.
- **All other code** (not static variables) **must be** inside the main method.
- The generation of random target locations and the engagement of the targets by the stormtrooper **must occur inside a SINGLE loop**. (*All output shown between the two “***” lines must occur inside a single loop*).
- A data structure is not required but the **only data structure** that can be used is a **1D Array**.
- The **determination of a hit or miss must be done** by generating a random number and making some comparison (*you decide*) to the stormtrooper's current accuracy...*you can make this process as sophisticated as you like*.

Expected Output

```
Problems @ Javadoc Declaration Console X
<terminated> Gene (4) [Java Application] C:\Users\msgth\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v20241024-1700\jre\bin\javaw.exe (Feb 2, 2025, 2:27:27 PM - 2:27:28 PM elapsed: 0:00:00.135) [pid: 59660]

                                Stormtrooper Marksmanship Results
Target  Tgt X    Tgt Y    Tgt Distance    Tgt Range    Tgt Quadrant    Hit-Miss    Trpr Accuracy
*****
1       34       7       11.40m         Medium       SE              Hit          70.0
2       18       12      13.89m         Medium       SW              Hit          70.0
3       32       15      16.55m         Medium       SE              Hit          70.0
4       36       35      36.69m         *OUT*        NE              No Shot      70.0
5       33       13      15.26m         Medium       SE              Miss         70.0
6       44       6       19.92m         Medium       SE              Miss         65.0
7       11       8       16.12m         Medium       SW              Hit          60.0
8       40       39      41.79m         *OUT*        NE              No Shot      60.0
9       29       6       7.21m          Short        SE              Hit          60.0
10      2        19      29.83m         Long         SW              Miss         60.0
*****
Stormtrooper Engaged: 8 targets
    Targets Hit: 5 Targets Missed: 3
    Targets Out of Range: 2
Trooper Overall Accuracy: 5 hits / 8 shots...62.50%
Average Target Distance Was: 20.87 meters
```

While your values will be different, the **FORMAT** of your output **MUST BE EXACTLY** as shown.